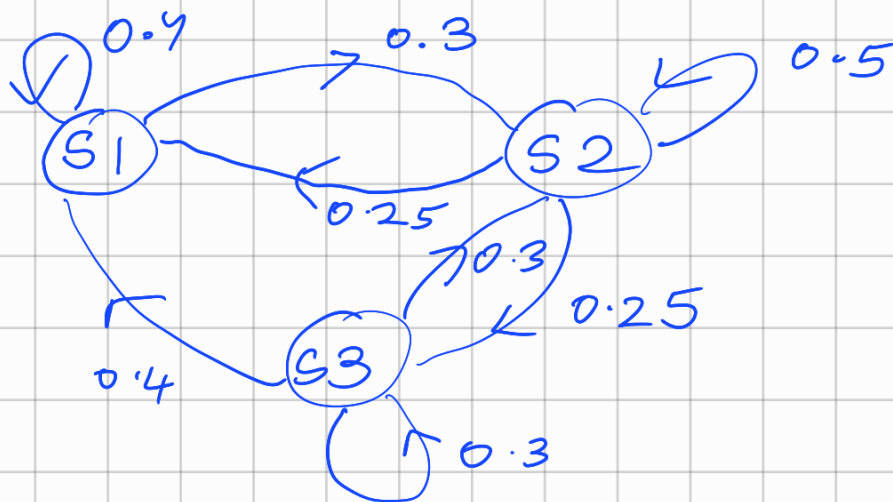


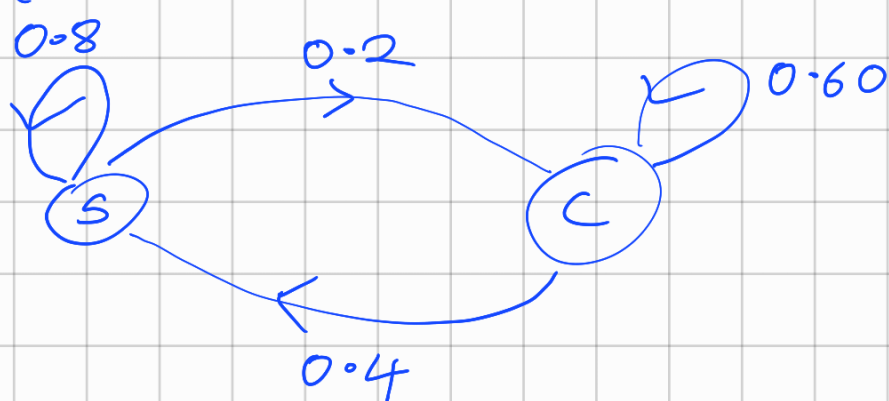
See Sheet 10 - Markov Chains

$$P = \begin{matrix} & \begin{matrix} s_1 & s_2 & s_3 \end{matrix} \\ \begin{matrix} s_1 \\ s_2 \\ s_3 \end{matrix} & \begin{bmatrix} 0.4 & 0.3 & 0 \\ 0.25 & 0.5 & 0.25 \\ 0.4 & 0.3 & 0.3 \end{bmatrix} \end{matrix}$$



Q3) $S = \{ \text{Sunny, Cloudy} \}$

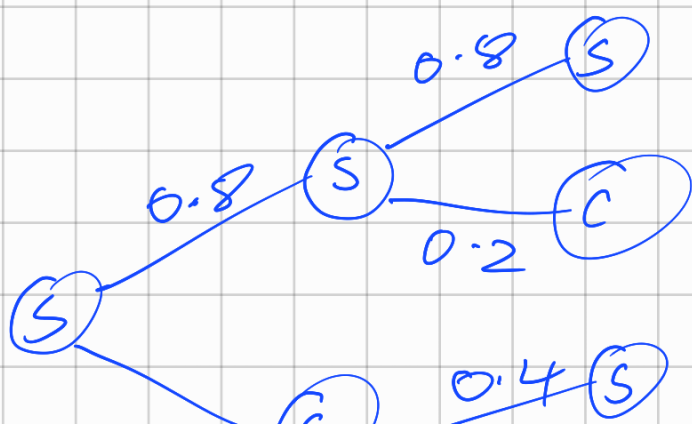
(a) (b)



$$P = \begin{matrix} & \begin{matrix} S & C \end{matrix} \\ \begin{matrix} S \\ C \end{matrix} & \begin{bmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{bmatrix} \end{matrix}$$

c) i) Prob. Tree
Tomorrow

Day after tomorrow



$$\dots (0.8)(0.2) = \underline{0.16}$$

$$0.2 \xrightarrow{\quad} \text{C} \xrightarrow{0.6} \text{C} \quad \dots (0.2)(0.6) = \underline{0.12}$$

$$P(\text{cloudy day after tomorrow}) = 0.16 + 0.12 = \underline{\underline{0.28}}$$

ii) Initial state vector

$$x(1) = \begin{bmatrix} S & C \\ 1 & 0 \end{bmatrix}$$

$$\left[\begin{array}{l} \text{Equally likely to} \\ \text{be sunny or} \\ \text{cloudy} \end{array} \right] x(1) = \begin{bmatrix} 0.5 & 0.5 \end{bmatrix}$$

Go Find
Future
states

$$x(m+1) = x(1) P^m$$

$$x(3) = x(1) P^2$$

$$= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{bmatrix}^2$$

$$= \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{bmatrix} \begin{bmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{bmatrix}$$

$\begin{bmatrix} S & C \end{bmatrix}$

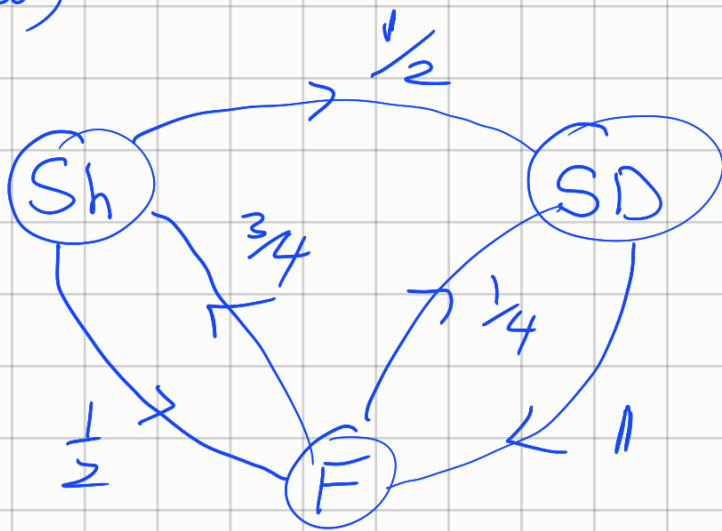
$$= [0.72 \quad 0.28]$$



Probability of it being cloudy the day after tomorrow given that it was initially sunny.

Q4) $S = \{ \text{Shaggy, (Sh), Scooby Doo, (SD), Fred } (F) \}$

(a)

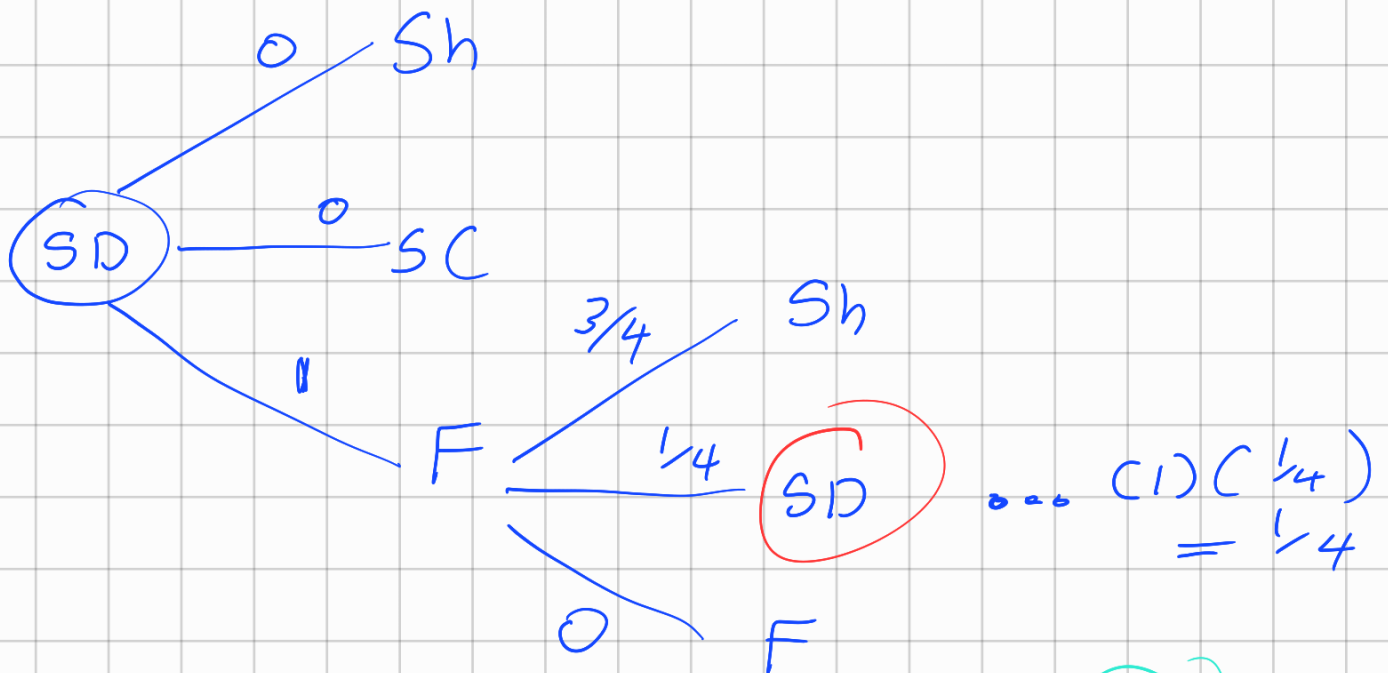


(b)

$$P = \begin{matrix} & \begin{matrix} Sh & SD & F \end{matrix} \\ \begin{matrix} Sh \\ SD \\ F \end{matrix} & \begin{bmatrix} 0 & 1/2 & 1/2 \\ 0 & 0 & 1 \\ 3/4 & 1/4 & 0 \end{bmatrix} \end{matrix}$$

(c)

1 throw : After 2 throws



$$P(\text{SD having frisbee after 2 throws}) = 1/4$$

at cor 2 throws
SD had Frisbee
initially.

Markov chains: $x(1) = \begin{matrix} & \text{Sh} & \text{SD} & \text{F} \\ \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \end{matrix}$

$$x(3) = x(1) P^2$$

$$= \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 \\ \frac{3}{4} & \frac{1}{4} & 0 \end{bmatrix}^2$$

$$= \begin{bmatrix} \frac{3}{4} & \frac{1}{4} & 0 \end{bmatrix}$$

Use Markov chain if for example
initially each person was equally likely
to have frisbee:

$$x(1) = \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}$$

d) long term probability (steady state)

$$x(101) = x(1) P^{100}$$

$$x(102) = x(1) P^{101}$$

Are the states the same?

$$x(102) = x(1) P^{101}$$

$$x(102) = x(101) P$$



Equal if in steady state

Solve $x = x P$

x = long term steady state prob.

$$x = [x_1 \quad x_2 \quad x_3]$$

$$x P = x$$

$$[x_1 \quad x_2 \quad x_3] \begin{bmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ 0 & 0 & 1 \\ \frac{3}{4} & \frac{1}{4} & 0 \end{bmatrix} = [x_1 \quad x_2 \quad x_3]$$

$$\begin{aligned} \textcircled{1} \quad \frac{3}{4}x_3 &= x_1 \\ \textcircled{2} \quad \frac{1}{2}x_1 + \frac{1}{4}x_3 &= x_2 \\ \textcircled{3} \quad \frac{1}{2}x_1 + x_2 &= x_3 \end{aligned}$$

$$\textcircled{1} \quad -x_1 + \frac{3}{4}x_3 = 0$$

$$\textcircled{2} \quad \frac{1}{2}x_1 - x_2 + \frac{1}{4}x_3 = 0$$

$$\textcircled{3} \quad \frac{1}{2}x_1 + x_2 - x_3 = 0$$

Homogeneous system

1) ~~$x_1 = x_2 = x_3 = 0$~~
(trivial solution)

2) Non-trivial solution
→ Infinite number of

$$\det \begin{pmatrix} -1 & 0 & 3/4 \\ 1/2 & -1 & 1/4 \\ 1/2 & 1 & -1 \end{pmatrix} = 0 \rightarrow \text{solution 5}$$

$$\left[\begin{array}{ccc|c} -1 & 0 & 3/4 & 0 \\ 1/2 & -1 & 1/4 & 0 \\ 1/2 & 1 & -1 & 0 \end{array} \right]$$

↓ Perform row operations to find

$$\left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 5/8 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right] \rightarrow \text{2 equations \& 3 unknowns}$$

Replace one of the original equations with following:

$$x_1 + x_2 + x_3 = 1$$

$$\rightarrow \left[\begin{array}{ccc|c} -1 & 0 & 3/4 & 0 \\ 1/2 & -1 & 3/4 & 0 \\ 1/2 & 1 & -1 & 0 \end{array} \right] \text{ Replace with } *$$

$$\left[\begin{array}{ccc|c} -1 & 0 & 3/4 & 0 \\ 1 & 1 & 1 & 1 \\ 1/2 & 1 & -1 & 0 \end{array} \right] \xrightarrow{R_1 = -1R_1} \text{Non-homogeneous system}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & x_1 \\ 0 & 1 & 0 & x_2 \\ 0 & 0 & 1 & x_3 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 3/4 & 0 \\ 1 & 1 & 1 & 1 \\ 1/2 & 1 & -1 & 0 \end{array} \right]$$

$$\begin{aligned} R_2 &= R_2 - R_1 \\ R_3 &= R_3 - \frac{1}{2}R_1 \end{aligned}$$

$$\begin{bmatrix} 1 & 0 & -3/4 & 0 \\ 0 & 1 & 1/4 & 1 \\ 0 & 1 & -5/8 & 0 \end{bmatrix}$$

$$R_3 = R_3 - R_2$$

$$\begin{bmatrix} 1 & 0 & -3/4 & 0 \\ 0 & 1 & 1/4 & 1 \\ 0 & 0 & -19/8 & -1 \end{bmatrix}$$

$$R_3 = -8/19 R_3$$

$$\begin{bmatrix} 1 & 0 & -3/4 & 0 \\ 0 & 1 & 1/4 & 1 \\ 0 & 0 & 1 & 8/19 \end{bmatrix}$$

$$\begin{aligned} R_1 &= R_1 + \frac{3}{4} R_3 \\ R_2 &= R_2 - \frac{1}{4} R_3 \\ x_3 &= 8/19 \end{aligned}$$

$$\begin{bmatrix} 1 & 0 & 0 & 6/19 \\ 0 & 1 & 0 & 5/19 \\ 0 & 0 & 1 & 8/19 \end{bmatrix}$$

Sh SD F

$$\underline{x} = [x_1 \ x_2 \ x_3] = \left[\begin{array}{ccc} 6/19 & 5/19 & 8/19 \end{array} \right]$$

Long term probability
that Shaggy will
have that Frisbee.

$x(11), x(101)$

